The Trading Theorem for SIC Formal Systems

Audited and repaired research edition v2.0

Brian Tenneson
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Abstract

This note formulates the trade between hypotheses and inference resources in SIC formal systems, with explicit distinctions among syntactic proof translation, equality of consequence closures, and state-by-state operational equivalence. Moving an assumed formula into a zero-premise rule is unconditional. Replacing a proper rule by an internalized formula is not: a two-sided, context-safe proof-simulation condition is needed. We give such a criterion, an instructive counterexample to a weaker criterion, an exact closure theorem, a careful treatment of induction and recursive induction, and conditions for transferring an attained SIC proof state between isomorphic presentations. No bound on search cost, operational equivalence, or implementation certification follows automatically.

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1 SIC formal-system notation and scope

A formal system is a triple $F = (x, y, z)$ consisting of an alphabet x, a set y of well-formed formulas (WFFs), and a stock z of finitary inference-rule instances. Here a zero-premise rule is allowed. For $\Gamma \subseteq y$, let

$$\mathrm{Con}_F(\Gamma) = \{\varphi \in y : \text{there is a finite } F\text{-derivation of } \varphi \text{ from } \Gamma\}.$$

Rule applications are required to meet their actual side conditions, including variable freshness, substitution, and proof support. An effective claim additionally requires a decidable presentation of the relevant rule instances and effectively generable finite proof templates.

A SIC state is written $C_F(\Gamma, m)$ for the formulas attained at stage m. Whenever we speak of *the* SIC closure, we explicitly assume

$$C_F(\Gamma, m) \subseteq C_F(\Gamma, m+1), \qquad \bigcup_{m \geq 0} C_F(\Gamma, m) = \mathrm{Con}_F(\Gamma). \tag{1}$$

The second equality is a condition on the chosen SIC scheduler; it is not true of every budgeted or unfair search procedure.

2 The unconditional basic trading theorem

Definition 2.1 (Zero-premise rule)**.** For $a \in y$, write $r_a = \dfrac{-}{a}$ for the rule that derives a with no premises.

Theorem 2.2 (Basic trading theorem)**.** *For any $F = (x, y, z)$, $\Gamma, A_0 \subseteq y$, and $F' = (x, y, z \cup \{r_a : a \in A_0\})$,*

$$\mathrm{Con}_F(\Gamma \cup A_0) = \mathrm{Con}_{F'}(\Gamma).$$

If the sets and proof records are effectively represented, the conversion of any fixed finite proof is effective.

Proof. Replace each initial use of an assumption $a \in A_0$ by the corresponding zero-premise inference r_a. Conversely, replace each such inference by a hypothesis leaf a. Other proof steps and their side conditions remain unchanged. In either direction the support change is precisely the declared trade between $\Gamma \cup A_0$ and Γ. □

Corollary 2.3 (Theorem-set invariance under a fixed bank)**.** *At $\Gamma = \varnothing$, adding A_0 as hypotheses or as zero-premise rules yields the same closed consequence set. This assertion does not claim equality with the original empty-bank theorem set* $\mathrm{Con}_F(\varnothing)$*.*

No conjunction, implication, quantifier, or special object-language connective is required.

3 Proper rules, internalization, and the missing reverse implication

Let $z = z_0 \cup R$ and $F_0 = (x, y, z_0)$. A proposed internalization of each $\rho \in R$ is a formula $b_\rho \in y$ (or a schema whose instances are explicitly listed). Let $B = \{b_\rho : \rho \in R\}$ and define $Q = (x, y, z_0)$ with initial resources $\Gamma \cup B$.

Definition 3.1 (Context-safe two-sided trade)**.** The family B is a *faithful contextual trade* for R relative to z_0 if:

(S) For every admissible application of any $\rho \in R$, with concrete premises $\varphi_1, \ldots, \varphi_k$, there is a finite Q-derivation of its conclusion from those premises and B, preserving the intended contextual judgments and their side conditions.

(R) Every added resource $b \in B$ has a finite F-derivation from the empty hypothesis bank, or (for a fixed-context result only) an F-derivation from the declared Γ.

If *effective* translation is asserted, the witnesses in (S) and (R) must be constructible from the corresponding finite instance descriptions.

Condition (S) is simulation of the eliminated rule. Condition (R) is recoverability of the new hypothesis or axiom leaf. Merely saying that "intended uses" of an internalizing formula simulate the old rule is insufficient when the new formula is available as an unrestricted assumption.

Theorem 3.2 (Repaired rule-to-hypothesis trading theorem)**.** *If condition (R) of Definition 3.1 is witnessed by empty-bank F-proofs for all members of B (and (S) holds), then for every admissible Γ,*

$$\mathrm{Con}_{(x,y,z_0\cup R)}(\Gamma) = \mathrm{Con}_{(x,y,z_0)}(\Gamma \cup B). \tag{2}$$

When (S) and (R) are effective, the two inclusions are witnessed by effective proof-producing translations. If (R) is known only from a particular fixed Γ, the displayed equality is asserted only for that fixed Γ. These translations need not be inverse on proof syntax.

Proof. For the left-to-right direction, traverse an F-proof in topological order. Leave hypotheses and z_0-steps unchanged; replace each occurrence of an eliminated rule with the finite simulation given by (S), importing exactly the formulas in B. The resulting Q-proof uses assumptions only from $\Gamma \cup B$. In the other direction, traverse a Q-proof, replacing each initial $b \in B$ with its F-proof given by (R); retain all z_0-steps. The result is an F-proof under Γ. The stated effectiveness follows from the effective templates. □

Remark 3.3 (Schematic and quantified internalizations). Writing $\widehat{\rho} = \forall \bar{v}\,((\varphi_1 \wedge \cdots \wedge \varphi_k) \to \psi)$ is meaningful only when $\bar{v}$ consists of actual object-language variables and the displayed expression is an actual WFF. Schematic metavariables and side conditions do not become object-language variables simply by writing $\forall$. Substitution, eigenvariable restrictions, and discharged-assumption scope require separate witnesses for (S) and (R). The added resource may have to be a *family* of formulas rather than a single sentence.

Counterexample 3.4 (Why one-way simulation fails)**.** *Start with classical propositional Hilbert logic (including modus ponens), and add the special rule p/q for two distinct propositional atoms p, q. A formula $p \to q$ would make this rule simulable by modus ponens, so the forward simulation (S) holds with $B = \{p \to q\}$. But $p \to q$ is* not *a theorem of the original system: in any closed derivation the rule cannot be applied for the first time, because before the first application all formulas are propositional tautologies and p is not. The new presentation* does *prove $p \to q$ from its added hypothesis. Thus the consequence sets differ at $\Gamma = \varnothing$. The missing condition is (R).*

Remark 3.5. This example also shows why equality of closed theorem sets alone is insufficient for arbitrary-context equivalence. An MP-only presentation with the usual implicational deduction theorem cannot in general represent the contextual step $\{p\} \vdash q$ while preserving the original closed theorem set.

4 General trading and proof translations

Theorem 4.1 (General contextual trading theorem)**.** *Let $F_1 = (x, y, z_1)$ and $F_2 = (x, y, z_2)$ share the same judgments. Suppose every finite proof resource needed by one presentation has a scope-correct finite simulation in the other presentation, in* both *directions, including initial hypotheses and any extra axiom leaves. Then*

$$\mathrm{Con}_{F_1}(\Gamma_1) = \mathrm{Con}_{F_2}(\Gamma_2).$$

If those simulations are algorithmic, there are total effective functions on finite proof encodings in both directions.

Proof. Induct over a finite derivation in each system. Translate each hypothesis leaf or rule occurrence using its specified simulation, retaining the side conditions and provenance. This produces the respective inclusion. Apply the argument again in the reverse direction. □

The theorem is a compositional criterion, not a claim that arbitrary rule/hypothesis trades necessarily satisfy its hypotheses. For example, the original trading principle is particularly straightforward for zero-premise rules by Theorem 2.2. The stronger first-order MP-only normalization in the companion manuscript *Trading, Intuition, and Return* relies on a distinct prefix/support construction; it should not be inferred merely from this abstract criterion.

5 SIC-state form and operational differences

Corollary 5.1 (Reachable-closure invariance)**.** *If both SICs satisfy Equation* (1) *and the general contextual trading hypotheses hold, then*

$$\bigcup_{m\geq 0} C_{F_1}(\Gamma_1, m) = \bigcup_{m\geq 0} C_{F_2}(\Gamma_2, m).$$

Proof. Each union equals its corresponding consequence closure by the SIC assumption; these consequence closures agree by the trading theorem. □

This does not imply $C_{F_1}(\Gamma_1, m) = C_{F_2}(\Gamma_2, m)$ for the same m: moving an assumption into a zero-premise rule can delay its acquisition by a stage. Proof length, runtime, memory, and cost are also not preserved.

6 Deductive isomorphisms and exact transport

Definition 6.1 (Deductive isomorphism)**.** A bijection $h : y \to y'$ between F and G is a deductive isomorphism if for every $\Gamma \subseteq y$, $\varphi \in y$,

$$\Gamma \vdash_F \varphi \quad \Longleftrightarrow \quad h[\Gamma] \vdash_G h(\varphi).$$

Theorem 6.2 (Consequence transport)**.** *For such h, $h[\mathrm{Con}_F(\Gamma)] = \mathrm{Con}_G(h[\Gamma])$.*

Proof. For each $\varphi \in y$, membership in the left side is equivalent to $\Gamma \vdash_F \varphi$, which by definition is equivalent to $h[\Gamma] \vdash_G h(\varphi)$, hence to membership in the right side. □

Definition 6.3 (SIC dynamical isomorphism)**.** A deductive isomorphism is *stage preserving* for specified SIC dynamics if $h[C_F(\Gamma, m)] = C_G(h[\Gamma], m)$ for every admissible Γ, m.

The latter is a strictly stronger operational requirement. If a full machine state carries budgets, cursors, proof receipts, or a current recursion value, a true operational isomorphism must transport those components and commute with executable transitions, not merely map attained formulas.

7 Induction as a traded deductive resource

Let $P(n)$ be a formula in an arithmetic language. Consider the rule

$$\rho_{\mathrm{Ind}} : \qquad \frac{P(0) \qquad \forall n(P(n) \to P(n+1))}{\forall n P(n)},$$

and the candidate internalization

$$I_P := (P(0) \wedge \forall n(P(n) \to P(n+1))) \to \forall n P(n).$$

Corollary 7.1 (Conditional induction trade)**.** *If the I_P family satisfies both (S) and (R) of Definition 3.1 relative to the remaining rules, then induction rule applications may be traded for the I_P family without changing contextual consequence.*

Proof. An instance of the induction rule is simulated from its two premises and I_P using conjunction introduction and MP; the reverse translation replaces each I_P-leaf with its assumed original proof. Invoke Theorem 3.2. □

Necessary qualification. Merely postulating an induction *rule* does not automatically prove the corresponding implication I_P as a closed theorem. Deriving I_P by assumption-discharge may itself require a deduction theorem compatible with the rule and all its side conditions. In systems where the induction *axiom schema* is already primitive, recoverability is immediate; otherwise it must be proved rather than inferred.

8 Recursive induction with well-formed data

The informal recursion $y_0 = a$, $y_{n+1} = f(n, y_n)$ is external notation until it is represented by terms or formulas in the language. For a genuine object-language statement, introduce a function symbol g with declared recurrence axioms

$$g(0) = a, \qquad g(n+1) = f(n, g(n)),$$

or a specified definable graph of the recursion. Then the well-formed induction target is $P(n, g(n))$, and its schema is

$$I_{P,g} := (P(0, g(0)) \wedge \forall n\,[P(n, g(n)) \to P(n+1, g(n+1))]) \to \forall n\, P(n, g(n)).$$

Trading this rule for $I_{P,g}$ is valid under the same two-sided criterion, plus the recurrence and typing assumptions. If the recursion is only a computation outside the language, the formula $\forall n P(n, y_n)$ is not justified as an ordinary WFF without an explicit encoding.

9 Induction handoff between SICs

Theorem 9.1 (State-aware induction handoff)**.** *Suppose h is a deductive isomorphism between F, G, a finite proof state $S_m = C_F(\Gamma, m)$ is given, and an attained formula $P(k, g(k)) \in S_m$ is certified. Assume further that (a) the translated recursive data and successor inference are available in G; (b) every future inference required for the transported induction is legal and supported in G; and (c) the handoff initializes the G-SIC with $h[S_m]$ and the required operational metadata. Then*

$$h(P(k, g(k))) \in C_G(h[S_m], 0),$$

and each subsequent legal induction successor can be translated and continued from that state without rederiving the initial segment.

Proof. The attained formula is in the newly supplied initial bank by construction. By (a)–(b) every specified successor step has a corresponding legal G-step; iterate over the desired finite number of successors. Assumption (c) ensures that no needed recursion value, authorization, or proof receipt is lost. □

Boundary. Deductive isomorphism *alone* does not imply stage-preserving handoff or sound transplantation of arbitrary operational histories. In particular the source's symbolic arrows $h(P(k, g(k))) \Rightarrow \cdots$ denote actual inference steps only when the successor mechanism and its hypotheses are supplied.

10 ATP interpretation and open directions

The repaired theorem separates eventual deductive reachability from the geometry, costs, and history of search. A controlled ATP experiment may move between verified proof states and faithfully translated presentations, then continue search with unchanged verification obligations. A trade can reorganize deductive resources without adding logical consequences, but it may change the order and size of the search frontier.

Open questions include characterization of context-faithful internalization for particular logics, quantitative proof-size distortion, SIC dynamical bisimulation, typed recursive induction handoff, and matched measurements of ATP search performance. None is settled by equality of consequence sets.

11 ATLAS v2.0 audit ledger

ID	Issue and action	Status
T01	Added recoverability (R) to correct the originally insufficient internalization hypothesis.	Repaired mathematically
T02	Added p/q counterexample to one-way rule simulation.	Repaired mathematically
T03	Distinguished hypothetical formula schema from genuine object-language universal closure.	Qualified
T04	Induction implication requires suitable deduction theorem or direct original derivability.	Conditional

T05	Replaced unbound y_n object-language notation with a represented recurrence $g(n)$.	Repaired
T06	Separated deductive, stagewise, and complete-machine isomorphisms.	Repaired
T07	No external proof assistant, live SIC engine, or performance experiment run for this edition.	Unverified externally

Release conclusion. The stated conditional theorems have explicit ordinary mathematical proofs; the manuscript is not accompanied by an independent machine-checked formalization. The absolute claim "no problems remain" cannot be certified for an open-ended mathematical manuscript.

References

1. B. Tenneson, *The Trading Theorem for SIC Formal Systems*, preliminary source draft, 5 November 2026 (the supplied source manuscript).

2. B. Tenneson, *Trading, Intuition, and Return: Certified Round-Trip Proof Translation for First-Order Theories and Automated Provers*, working paper 0.1, 26 October 2026 (supplied companion manuscript).

3. B. Tenneson, *Trading Theorem 003: Mutual Reproducibility, Consequence-Preserving Trades, Quotient Hunting, and Verified Settlement Search*, September 2026; author publication library.

4. Standard background: the Hilbert deduction theorem, first-order quantifier side conditions, and ordinary finitary consequence closure. These background principles are not new assertions of this manuscript.

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